NETWORK TRAFFIC INVESTIGATION

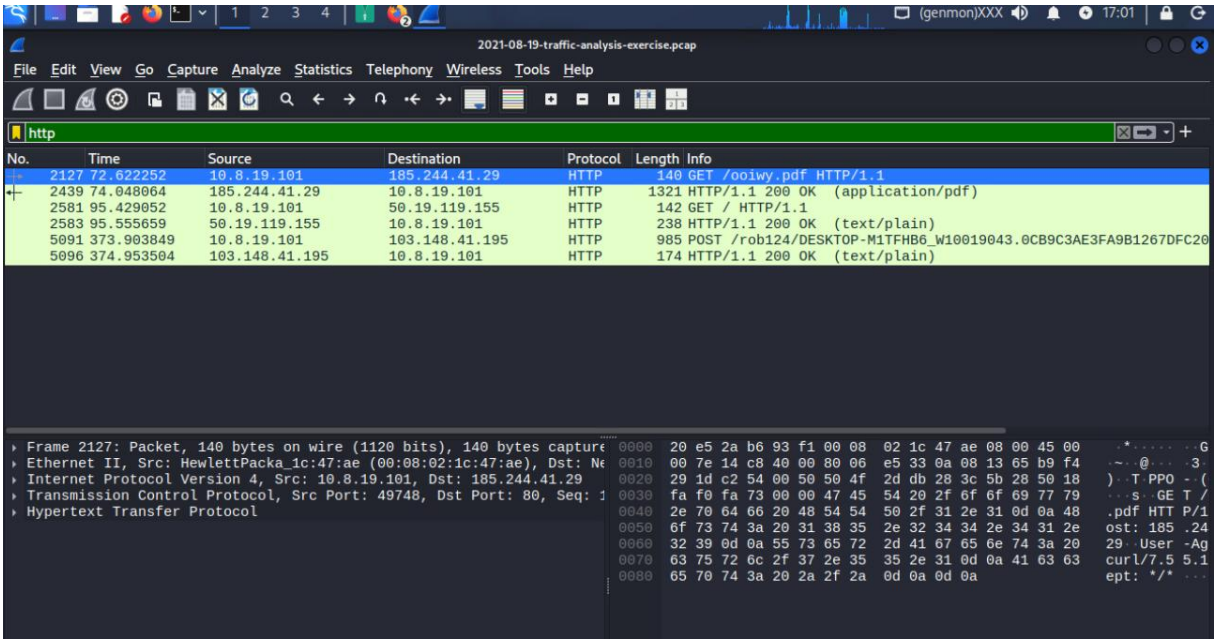
HTTP PACKET ANALYSIS WITH WIRESHARK

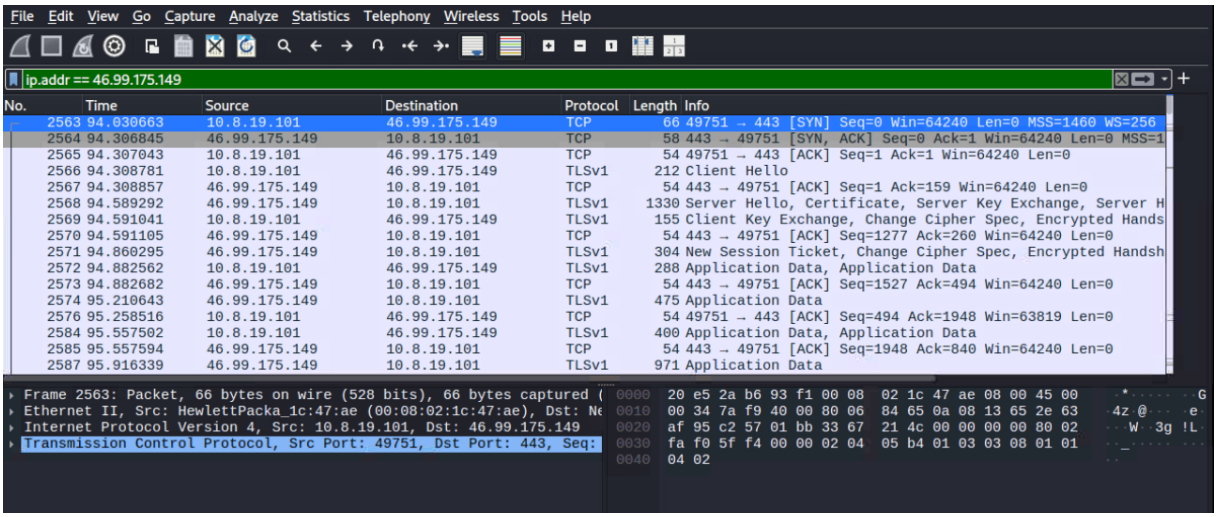
Contexte:

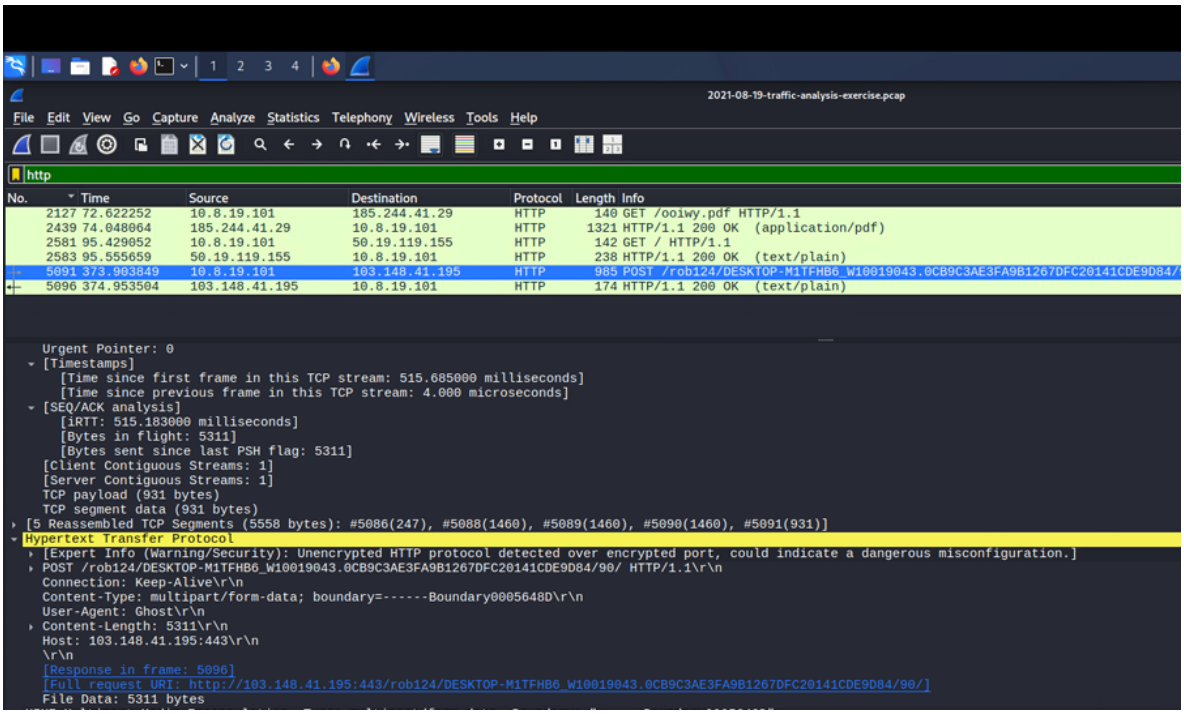
Traffic Analysis • HTTP Inspection • Packet Investigation

Introduction :

Analyse d'un trafic HTTP non chiffré à l'aide de Wireshark afin d'identifier des échanges sensibles, comprendre la structure des requêtes et observer les risques liés à des protocoles non sécurisés.


2021-08-19-traffic-analysis-exercise.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http
No. Time Source Destination Protocol Length Info
2127 72.622252 10.8.19.101 185.244.41.29 HTTP 140 GET /ooiwy.pdf HTTP/1.1
2439 74.048064 185.244.41.29 10.8.19.101 HTTP 1321 HTTP/1.1 200 OK (application/pdf)
2581 95.429052 10.8.19.101 50.19.119.155 HTTP 142 GET / HTTP/1.1
2583 95.555659 50.19.119.155 10.8.19.101 HTTP 238 HTTP/1.1 200 OK (text/plain)
5091 373.903849 10.8.19.101 103.148.41.195 HTTP 985 POST /rob124/DESKTOP-M1TFHB6_W10019043.0CB9C3AE3FA9B1267DFC20
5096 374.953504 103.148.41.195 10.8.19.101 HTTP 174 HTTP/1.1 200 OK (text/plain)
Frame 2127: Packet, 140 bytes on wire (1120 bits), 140 bytes capture
Ethernet II, Src: HewlettPacka_1c:47:ae (00:08:02:1c:47:ae), Dst: Ne
Internet Protocol Version 4, Src: 10.8.19.101, Dst: 185.244.41.29
Transmission Control Protocol, Src Port: 49748, Dst Port: 80, Seq: 1
Hypertext Transfer Protocol


File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
ip.addr == 46.99.175.149
No. Time Source Destination Protocol Length Info
2563 94.030663 10.8.19.101 46.99.175.149 TCP 66 49751 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256
2564 94.306845 46.99.175.149 10.8.19.101 TCP 58 443 → 49751 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1
2565 94.307043 10.8.19.101 46.99.175.149 TCP 54 49751 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
2566 94.308781 10.8.19.101 46.99.175.149 TLSv1 212 Client Hello
2567 94.308857 46.99.175.149 10.8.19.101 TCP 54 443 → 49751 [ACK] Seq=1 Ack=159 Win=64240 Len=0
2568 94.589292 46.99.175.149 10.8.19.101 TLSv1 1330 Server Hello, Certificate, Server Key Exchange, Server H
2569 94.591041 10.8.19.101 46.99.175.149 TLSv1 155 Client Key Exchange, Change Cipher Spec, Encrypted Hands
2570 94.591105 46.99.175.149 10.8.19.101 TCP 54 443 → 49751 [ACK] Seq=1277 Ack=260 Win=64240 Len=0
2571 94.860295 46.99.175.149 10.8.19.101 TLSv1 304 New Session Ticket, Change Cipher Spec, Encrypted Handsh
2572 94.882562 10.8.19.101 46.99.175.149 TLSv1 288 Application Data, Application Data
2573 94.882682 46.99.175.149 10.8.19.101 TCP 54 443 → 49751 [ACK] Seq=1527 Ack=494 Win=64240 Len=0
2574 95.210643 46.99.175.149 10.8.19.101 TLSv1 475 Application Data
2576 95.258516 10.8.19.101 46.99.175.149 TCP 54 49751 → 443 [ACK] Seq=494 Ack=1948 Win=63819 Len=0
2584 95.557502 10.8.19.101 46.99.175.149 TLSv1 400 Application Data, Application Data
2585 95.557594 46.99.175.149 10.8.19.101 TCP 54 443 → 49751 [ACK] Seq=1948 Ack=840 Win=64240 Len=0
2587 95.916339 46.99.175.149 10.8.19.101 TLSv1 971 Application Data
Frame 2563: Packet, 66 bytes on wire (528 bits), 66 bytes captured (
Ethernet II, Src: HewlettPacka_1c:47:ae (00:08:02:1c:47:ae), Dst: Ne
Internet Protocol Version 4, Src: 10.8.19.101, Dst: 46.99.175.149
Transmission Control Protocol, Src Port: 49751, Dst Port: 443, Seq:



Observations Techniques

-Trafic HTTP transmis en clair

-Requête POST détectée lors de l'inspection des paquets

-Exposition potentielle d'informations sensibles

-Absence de chiffrement TLS observée

-Les communications réseau non chiffrées augmentent les risques d'interception

Protocole observé : HTTP
Outil d'analyse : Wireshark
Type de trafic : Communication web non chiffrée

Observation d'une requête POST transitant en HTTP sur le port 80.

Le trafic n’étant pas chiffré, les données peuvent être inspectées directement dans les paquets réseau.

Le protocole HTTP transmet les données en clair.

Une mauvaise configuration peut exposer des informations sensibles lors du transit réseau.

Conclusion :

Cette analyse permet de comprendre les risques liés aux protocoles non sécurisés ainsi que l’importance du chiffrement TLS dans les infrastructures modernes.

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